VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
FACULTY OF GEOLOGY AND PETROLEUM ENGINEERING

FIELD TRIP REPORT

EARTH SCIENCE – GE1013

TRIP 1: HO CHI MINH CITY – VUNG TAU – PHAN THIET
(30 OCTOBER – 1 NOVEMBER 2026)

Class: A01_A02 Group: 35

Advisor: Dr. Nguyễn Ngọc Thủy

Student Name	Student ID	Contribution
Hồ Công Thành	2313111	100%
Nguyễn Đình Huy	2311175	100%
Nguyễn Thành Nhân	2312440	100%
Thái Khắc Hùng	2311306	100%
Cao Chí Nguyên	2312331	100%

Ho Chi Minh City – 2026

Contents

List of Figures **3**

List of Tables **4**

1 Introduction **5**
1.1 Purpose of Preparing the Field Trip Report 5
1.2 Task Assignment Table of Group Members 5
1.3 Natural, Technical, and Socio-economic Characteristics of the Study Area 5
1.4 Tentative Schedule 6
1.4.1 Planned Route and Fieldtrip Photograph 6
1.4.2 Field Activities and Documentation 6

2 Geosphere **7**
2.1 Weathering in Laterite and Formation Mechanism 7
2.1.1 Chemical Weathering and Leaching Mechanisms 8
2.1.2 Field Observations at the Ho Da Outcrop 8
2.2 Erosion and Coastal Processes 10
2.2.1 Marine Wave Action and Beach Sediment Sorting at Bai Sau 10
2.2.2 Coastal Erosion and Seawall Protection at Loc An 11
2.2.3 Aeolian Dune Transport and Wind Ripple Bedforms 12
2.3 Magma Eruption Processes and Rhyolite 13
2.3.1 Textural Classification Based on Gas Degassing and Structure 13
2.3.2 Silicate Rocks versus Carbonate Rocks 16
2.3.3 Rhyolite of the at Bai Sau (Vung Tau) 16
2.4 Granitic Intrusion, Diabase Dykes, and Cross-cutting Relationships 17
2.4.1 Granitic Intrusion 18
2.4.2 Diabase Dykes 18
2.4.3 Cross-cutting Relationship of Intrusive Igneous Rocks 19
2.4.4 Height Estimation of the Red Sand Escarpment Using a Human Scale 20
2.4.5 Geomorphological and Slope Stability Implications 22
2.5 Measuring the Orientation of Fractures with a Geological Compass 22
2.5.1 Measurements at Bai Sau (Vung Tau) 22
2.5.2 Measurements at Mui Ke Ga 23
2.5.3 Comparison and Geological Significance 23
2.6 Summary of Geosphere Observations 24

3 Hydrosphere **25**
3.1 Surface Water and Stream Processes at Suoi Tien (Fairy Stream) 25
3.2 Groundwater Dynamics in Coastal Sand Dune Systems (Ho Tram & Coastal Dunes) . . . 25
3.3 Coastal Erosion, Wave Action, and Marine Processes (Back Beach, Mui Ke Ga, & Ho Tram Estuary) 26
3.4 Impact of Mining Pit Water Retention at Ho Da and Gia Quy Quarry 26

4 Biosphere **28**
4.1 Pioneer Vegetation and Biological Weathering at Ho Da (VNU-HCM Lake) 28
4.2 Plant Colonization of Basalt and Pozzolan Surfaces at Gia Quy Quarry 28
4.3 Coastal Vegetation at Bai Sau and Cua Ho Tram – Loc An 29
4.4 Vegetation on the Rocky Headland at Mui Ke Ga 32
4.5 Vegetation and Slope Stability at Suoi Tien (Fairy Stream) 33
4.6 Summary of Biosphere Observations 33

5 Atmosphere **33**
5.1 Climatic Characteristics of Vung Tau 35
5.2 Climatic Characteristics of Phan Thiet and Their Landscape Significance 36

6 Natural Resources **37**
6.1 Construction Stone and the Post-mining Landscape at Ho Da 37
6.2 Construction Stone Resources and Quarry Engineering at Gia Quy 38
6.3 Petroleum Resources and Offshore Technology at Vietsovpetro 40
6.3.1 Offshore Structures and Their Engineering Role 41
6.3.2 Drilling Technology and Economic Efficiency 41

7 Conclusion **42**

References **44**

List of Figures

1 GE1013 fieldtrip participants (Vehicle 4) . . . 6
2 Laterite weathering profile at Ho Da . . . 9
3 Nodular laterite texture at Ho Da . . . 10
4 Beach sediment sorting and wave action . . . 11
5 Coastal sand erosion and deflation . . . 12
6 Asymmetric wind ripples in coastal sand . . . 13
7 Massive volcanic rock at Gia Quy Quarry . . . 14
8 Vesicular volcanic rock at Gia Quy Quarry . . . 15
9 Calcite specimen at Gia Quy Quarry . . . 16
10 Weathered rhyolite at the coast . . . 17
11 Diabase dyke and host rock at Mui Ke Ga . . . 19
12 Cross-cutting relationship at Mui Ke Ga . . . 20
13 Red and white sand escarpment at Suoi Tien . . . 21
14 Compass measurements of a fracture at Bai Sau . . . 22
15 Compass measurements of a fracture at Mui Ke Ga . . . 23
16 Stream erosion at Suoi Tien . . . 25
17 Coastal sand dunes and groundwater recharge . . . 26
18 Coastal erosion and wave action . . . 27
19 Waves acting on granite at Mui Ke Ga . . . 27
20 Water accumulation at Ho Da Quarry . . . 28
21 Vegetation colonising the Ho Da quarry rim . . . 29
22 Pioneer plants in Gia Quy Quarry . . . 30
23 Grass growing on Gia Quy Quarry slopes . . . 31
24 Coastal vegetation at Bai Sau . . . 31
25 Creeping vegetation and shrubs at Loc An . . . 32
26 Vegetation on Mui Ke Ga headland . . . 32
27 Vegetation on sand cliffs at Suoi Tien . . . 33
28 Fallen tree and slope erosion at Suoi Tien . . . 34
29 Atmospheric conditions at Bai Sau . . . 36
30 Atmospheric conditions at Mui Ke Ga . . . 37
31 Post-mining landscape at Ho Da . . . 38
32 Exposed quarry face at Gia Quy . . . 39
33 Geological samples and facilities at Gia Quy . . . 39
34 Offshore-field model at Vietsovpetro . . . 40
35 TAM DAO 02 and offshore platform models . . . 41

List of Tables

1 Task assignments of Group 35 5
2 Planned GE1013 fieldtrip itinerary (eight stops) 7
3 Compass measurements of fracture planes 23
4 Summary of geosphere field observations 24
5 Summary of biosphere field observations 35

1 Introduction

1.1 Purpose of Preparing the Field Trip Report

An outcrop, a beach, and an industrial facility may appear to be very different places, yet each offers evidence of how the Earth works and how people use its resources. The GE1013 fieldtrip gave our group a chance to examine these connections beyond the classroom. Over three days, we travelled from the university area of Ho Chi Minh City through the Vung Tau coast and onward to the Phan Thiet area, recording what we could observe at each stop.

We prepared this report as a record of that work. Its purpose is to bring together our photographs, compass measurements, specimen observations, and notes taken during the lecturers' explanations, and then use this material to describe the processes encountered in the field. We are particularly interested in the contrast between fresh and weathered rock, the movement of sand by water and wind, the shaping of coastal landscapes, and the use of geological knowledge in quarrying and petroleum operations.

The report also provides an opportunity to reconsider our first impressions after the trip. A colour, a fracture, or a landform is an observation; explaining how it formed requires a reasoned interpretation. We therefore aim to describe the evidence clearly and relate it to the concepts studied in Earth Science, while recognising that a short visit cannot answer every geological question. Just as importantly, preparing a shared record allows us to combine the different contributions made by each group member.

1.2 Task Assignment Table of Group Members

To document the trip from different perspectives, the members of Group 35 divided the field responsibilities among photography, measurements, specimen collection, and note-taking. The following assignments reflect each member's main role during the fieldtrip; members also assisted one another whenever needed.

Table 1: Task assignments of Group 35. Field responsibilities during the trip.

No.	Name	Main Tasks
1	Hồ Công Thành	Photographing and filming the activities and observations throughout the fieldtrip.
2	Nguyễn Đình Huy	Measuring geological specimens and features using a geological compass.
3	Nguyễn Thành Nhân	Accompanying the lecturer and recording essential explanations and observations.
4	Thái Khắc Hùng	Accompanying the lecturer and collecting geological specimens.
5	Cao Chí Nguyên	Accompanying the lecturer, collecting specimens, and taking field notes.

1.3 Natural, Technical, and Socio-economic Characteristics of the Study Area

The fieldtrip crossed several contrasting environments rather than focusing on a single geological exposure. At Ho Da in the VNU-HCM area, a former quarry provides access to bedrock, weathered material, and a water-filled excavation. Along the Vung Tau and Ho Tram coasts, waves, currents, and wind continually modify the sand and the shoreline. Farther along the route, the rocky headland at Mui Ke Ga contrasts with the sand dunes and the stream-cut slopes at Suoi Tien. Seeing these settings within one journey made differences in rock type, surface processes, and landscape development easier to recognise.

The places we visited also required different approaches to observation. At rocky sites, we looked for mineral characteristics, fractures, weathering features, and relationships between rock bodies. At beaches

and sandy areas, we paid closer attention to grain size, sediment movement, and the shapes produced by wind or flowing water. The geological compass, collected samples, photographs, and written notes helped us document details that could otherwise be missed during a brief stop.

Human activities formed another important part of the route. Gia Quy Quarry illustrates how geological materials are extracted and used, while the visit to Vietsovpetro introduced the engineering and operational side of offshore petroleum development. These stops encouraged us to consider not only the origin and properties of geological materials but also the practical decisions, environmental changes, and technical work associated with using them. Together, the sites provided a varied setting for our study of the geosphere, hydrosphere, atmosphere, biosphere, and natural resources.

1.4 Tentative Schedule

The GE1013 fieldtrip took place from 30 October to 1 November 2026. The journey was arranged to introduce the group to volcanic and intrusive rocks, coastal and wind-driven sedimentary processes, stream erosion, and the industrial applications of geology. The planned sequence of eight destinations is presented below in the order supplied for the trip; no separate arrival time is assigned where it has not been confirmed.

1.4.1 Planned Route and Fieldtrip Photograph

Figure 1: GE1013 fieldtrip participants (Vehicle 4).

The itinerary began near the university in Ho Chi Minh City, continued through the Vung Tau and Ho Tram area, and ended at coastal and sandy landscapes farther east. The order of field stops was as follows:

1.4.2 Field Activities and Documentation

The aim at each destination was not simply to collect a rock sample or take a photograph. We first listened to the lecturer's explanation, identified the main feature to be examined, and then recorded the evidence available at the site. Where appropriate, group members used a geological compass, noted the appearance of rocks and sediments, collected permitted specimens, and took photographs and video for later review.

The route gave us several opportunities to compare processes across locations. Rock exposures at Ho Da, Gia Quy, and Mui Ke Ga could be examined in terms of origin, texture, fractures, and weathering. Bai Sau and Ho Tram provided coastal examples, whereas the sand dunes and Suoi Tien allowed us

Table 2: Planned GE1013 fieldtrip itinerary (eight stops). Sequence of stops and main observation themes.

Stop	Destination	Main observation themes
1	Ho Da (VNU-HCM, University Village)	Exposed rock, lateritic weathering, and the former quarry landscape.
2	Back Beach (Bai Sau), Vung Tau	Beach sediments, wave action, and coastal sediment transport.
3	Vietsovpetro	Petroleum geology, offshore operations, and applications of geological knowledge.
4	Gia Quy Quarry	Basalt, volcanic textures, weathering, and construction-material extraction.
5	Ho Tram – Ba Ria	Coastal erosion and deposition, beach protection, and sandy environments.
6	Mui Ke Ga	Granite, fractures, intrusive relationships, and wave action on rocky shores.
7	Sand dunes	Wind-driven sediment transport and dune landforms.
8	Suoi Tien (Fairy Stream)	Layered sand deposits, stream erosion, and the development of steep sandy slopes.

to consider the different effects of wind and running water on loose sediments. Vietsovpetro added an industry-focused perspective to the geological observations.

After the trip, our group brought the records together, checked the descriptions against the available photographs and notes, and organised the findings by Earth Science topic. Figure 1 preserves a moment from the fieldtrip, while Table 2 summarises the intended sequence of stops used to structure our observations.

2 Geosphere

The geosphere encompasses the solid Earth, including rocks, minerals, landforms, and the dynamic geological processes that create and reshape the Earth's crust over time. During the Earth Science fieldtrip, geological observations focused on two major categories of geological processes:

1. **Endogenous Processes (Internal):** Magmatic activities, including deep plutonic intrusion of granite, shallow injection of diabase dykes, and extrusive volcanic eruptions of andesite, basalt, and rhyolite.

2. **Exogenous Processes (External):** Tropical chemical weathering (lateritization), wave-induced coastal erosion, littoral sediment sorting, and wind (aeolian) transport along coastal dunes.

This chapter presents field observations and geological interpretations across the required core topics, connecting real outcrop evidence with fundamental Earth Science principles.

2.1 Weathering in Laterite and Formation Mechanism

Laterite, locally known in Vietnam as "đá ong", is a strongly weathered residual soil and rock material characteristic of warm, humid tropical and subtropical climates with distinct wet and dry seasons. Laterite formation is driven by prolonged and intense chemical weathering combined with deep leaching by infiltrating rainwater.

2.1.1 Chemical Weathering and Leaching Mechanisms

Under tropical conditions, abundant rainfall charged with dissolved carbon dioxide forms weak carbonic acid (H_2CO_3). As this warm, acidic groundwater percolates through the parent rock over thousands to millions of years, primary silicate minerals undergo progressive chemical breakdown:

1. **Leaching of Soluble Elements:** Soluble carbonate minerals (such as calcite) dissolve and wash away first. Subsequently, mobile alkaline and alkaline-earth cations (Na^+, K^+, Ca^{2+}, Mg^{2+}) are leached out of silicate minerals, followed by progressive desilication (removal of soluble silica, SiO_2).

2. **Residual Concentration:** Insoluble sesquioxides of iron (Fe_2O_3, mainly hematite and goethite) and aluminum (Al_2O_3, mainly gibbsite) remain behind, becoming strongly concentrated in the upper weathering profile.

The primary chemical reaction involves the hydrolysis of feldspar minerals into secondary clay minerals (kaolinite), followed by further desilication into gibbsite:

$$2KAlSi_3O_8 \text{ (K-feldspar)} + 2H^+ + 9H_2O \longrightarrow Al_2Si_2O_5(OH)_4 \text{ (Kaolinite)} + 4H_4SiO_4 \text{ (aq)} + 2K^+ \quad (1)$$

$$Al_2Si_2O_5(OH)_4 + 5H_2O \longrightarrow 2Al(OH)_3 \text{ (Gibbsite)} + 2H_4SiO_4 \text{ (aq)} \quad (2)$$

Simultaneously, iron released from ferromagnesian minerals oxidizes in the aerated unsaturated zone to produce insoluble ferric oxides and hydroxides:

$$4Fe^{2+} + O_2 + 4H_2O \longrightarrow 2Fe_2O_3 \text{ (Hematite)} + 8H^+ \quad (3)$$

$$Fe^{3+} + 2H_2O \longrightarrow FeO(OH) \text{ (Goethite)} + 3H^+ \quad (4)$$

Hematite imparts a characteristic blood-red to reddish-brown coloration, while goethite produces yellowish-brown hues. When exposed to the atmosphere and dried, the accumulated iron oxides act as a natural cementing agent, hardening the material into a cellular, durable crust known as a ferricrete or duricrust.

2.1.2 Field Observations at the Ho Da Outcrop

At Stop 1 (Ho Da, VNU-HCM Quarry Lake), students observed an in-situ laterite profile developed directly over an intermediate volcanic rock basement (andesite). This stratigraphic relationship proves that the laterite formed in situ (in place) by weathering of the underlying volcanic bedrock.

Figure 2: Laterite weathering profile at Ho Da. Mottled reddish-brown and yellowish tones show residual iron and aluminum accumulation.

The outcrop displays key transitional weathering characteristics:

- **Bedding Attitude:** The contact between the weathered mantle and the underlying andesitic rock dips gently at approximately 10°–20°.
- **Texture and Structure:** The material displays a porous, cellular, and mottled texture with numerous visible voids left by dissolved minerals.
- **Weathering Maturity:** The upper zone is crumbly and friable, indicating a transitional stage of lateritization where iron cementation is ongoing but not yet fully consolidated into hard, mature rock. Resistant quartz grains remain scattered within the reddish clay matrix.

Figure 3: Nodular laterite texture at Ho Da. The close-up shows a rough surface, cellular pores, and concentrated iron-rich patches.

2.2 Erosion and Coastal Processes

Coastal zones represent dynamic boundaries where the geosphere constantly interacts with marine waves, tides, currents, and atmospheric winds. During the fieldtrip, coastal processes were examined at two contrasting sites: the open sandy beach of Bai Sau (Vung Tau) and the tidal inlet / dune system at Cua Ho Tram – Loc An.

2.2.1 Marine Wave Action and Beach Sediment Sorting at Bai Sau

At Bai Sau (Vung Tau), sedimentation is controlled by the balance between wave uprush (swash) and gravitational return flow (backwash):

- Incoming breaking waves suspend bottom sand grains and wash them up the foreshore slope.
- As swash water infiltrates into the permeable sand, backwash volume and velocity decrease, winnowing away fine, light particles while depositing clean, well-sorted sand.

- **Mineral Composition:** Sand samples consist primarily of resistant, sub-rounded monocrystalline quartz ($> 75\%$), with subordinate feldspar and bioclastic shell debris.
- **Ilmenite Heavy Mineral Placers:** Noticeable black sand streaks occur along the upper swash berm. These black grains are dense heavy minerals, predominantly ilmenite ($FeTiO_3$, density ~ 4.7 g/cm^3), with minor magnetite and rutile. Hydraulic density sorting by waves winnows away lighter quartz (density 2.65 g/cm^3), concentrating ilmenite into economically valuable titanium placer deposits.

Figure 4: Beach sediment sorting and wave action. The active swash zone and beach berm illustrate wave-reworked sediment sorting and littoral transport.

2.2.2 Coastal Erosion and Seawall Protection at Loc An

At Cua Ho Tram – Loc An, the coast experiences ongoing shoreline retreat due to intense wave attack and longshore drift:

- High-energy storm waves erode the beach face, creating steep erosional scarps in coastal sand deposits.
- In response, a rip-rap rock seawall (bờ kè Lộc An) has been constructed from stacked armor stones. The interlocking stones and open voids dissipate wave energy, preventing further shoreline retreat.

Figure 5: Coastal sand erosion and deflation. An eroded beach scarp and wind-deflated sand surface show the interaction between wave erosion and aeolian transport.

2.2.3 Aeolian Dune Transport and Wind Ripple Bedforms

Landward of the high-tide line, onshore winds pick up dry sand through saltation (bouncing), suspension, and surface creep, forming coastal dunes. The sand surface displays distinct asymmetrical wind ripples:

- Wind ripples exhibit a gentle upwind stoss slope (8°–10°) and a steep downwind lee slope ($\sim 30^\circ$).
- Ripple crests align strictly perpendicular to the dominant onshore wind direction, serving as reliable paleo-wind indicators.

Figure 6: Asymmetric wind ripples in coastal sand. Sharp crestlines are aligned perpendicular to the onshore wind direction.

2.3 Magma Eruption Processes and Rhyolite

Magma eruption, or volcanism, occurs when molten rock ascends from the mantle or lower crust and breaches the Earth's surface, cooling rapidly in contact with the atmosphere or seawater to form extrusive (volcanic) igneous rocks.

2.3.1 Textural Classification Based on Gas Degassing and Structure

During volcanic eruptions, magma undergoes sudden decompression as it rises from high-pressure deep magma chambers to atmospheric pressure. The behavior of dissolved volatile gases (H_2O, CO_2, SO_2) and the cooling rate determine the structural texture of the resulting volcanic rocks. Based on porosity, density, and degree of gas retention, volcanic igneous rocks are classified into three distinct physical categories:

1. **Dense / Massive Volcanic Rock**

Figure 7: Massive volcanic rock at Gia Quy Quarry.

- **Formation Mechanism:** Crystallizes in the slow-cooling interior or central core of thick lava flows, where high confining pressure and overburden weight suppress the expansion of dissolved gas bubbles.
- **Physical Characteristics:** Compact, homogeneous, and dense with virtually no visible voids or pores. Fresh specimens exhibit a pristine dark-grey to black color (in basalt) or greenish-grey color (in andesite/rhyolite), with an interlocking microcrystalline to aphanitic groundmass.
- **Engineering Value:** Due to its lack of internal porosity, fresh massive rock possesses exceptional compressive strength ($>$ 120 MPa), low water absorption, and high durability against abrasion. In the field, massive basalt from the Xuan Loc formation (at Gia Quy Quarry) and massive andesite (at Ho Da) serve as primary industrial crushed stone for high-strength concrete and asphalt road foundations.

2. **Vesicular Volcanic Rock**

Figure 8: Vesicular volcanic rock at Gia Quy Quarry.

- **Formation Mechanism:** Forms along the upper crust, margins, or advancing fronts of lava flows. As magma approaches the surface, rapid decompression causes dissolved volatile gases to exsolve and expand rapidly, forming discrete gas bubbles. When the lava crust chills rapidly, these bubbles are frozen in place before they can escape.
- **Physical Characteristics:** Contains abundant isolated spherical, elliptical, or tubular cavities called vesicles (lỗ rỗng khí), typically ranging from 1 mm to over 10 mm in diameter. The rock is notably lighter in density than massive basalt.
- **Secondary Mineralization (Amygdules):** In older or weathered lava flows, groundwater percolating through the porous rock deposits secondary minerals—such as white calcite ($CaCO_3$), microcrystalline quartz, or zeolites—within the vesicles, transforming them into filled cavities called amygdules (cấu tạo hạnh nhân).

3. **Scoriaceous / Frothy Volcanic Rock**

- **Formation Mechanism:** Originates during violent, explosive volcanic degassing (such as Strombolian-type eruptions) or at the frothy, agitated surface of gas-rich lava fountains. Volatiles expand explosively, frothing the molten magma into a sponge-like mass that quenches instantly in air.
- **Physical Characteristics:** Characterized by extremely high vesicularity and interconnected voids, often exceeding 50–80% of the total rock volume. In mafic compositions, this pro-

duces dark-colored, highly porous scoria (xỉ núi lửa), whereas in felsic, high-silica eruptions, it creates light-colored pumice (đá bọt) with walls so thin and density so low that it can float on water.

- **Pozzolanic Applications:** At Gia Quy Quarry, deposits of scoriaceous basalt, volcanic scoria, and pyroclastic ash tuffs form valuable pozzolan resources. The porous, amorphous volcanic glass reacts with lime and water to form hydraulic cementing compounds used extensively in civil engineering.

2.3.2 Silicate Rocks versus Carbonate Rocks

Rocks are classified by the dominant mineral group they contain:

- **Silicate rocks (basalt, andesite, rhyolite, granite):** Composed mainly of silicate minerals built from SiO_4 tetrahedra, such as quartz, feldspar, and biotite. Their silica (SiO_2) content controls magma viscosity and eruption style. These rocks do not react with dilute HCl.
- **Carbonate rocks (limestone, marble):** Composed mainly of calcite ($CaCO_3$), a carbonate mineral. They form mostly by chemical or biological precipitation in water, not directly from magma. Calcite fizzles strongly with dilute HCl, which is a quick field test to distinguish carbonate rocks from silicate rocks.

Figure 9: Calcite specimen at Gia Quy Quarry. The specimen was examined during the quarry visit.

All igneous rocks observed during this fieldtrip (andesite at Ho Da, basalt at Gia Quy Quarry, and granite at Ho Tram) are silicate rocks.

2.3.3 Rhyolite of the at Bai Sau (Vung Tau)

At coastal outcrops around Bai Sau (Vung Tau), students observed rhyolite belonging to the Late Cretaceous:

- **Petrology and Texture:** Fine-grained (aphanitic) to microcrystalline volcanic texture, reflecting rapid surface chilling. The mineral composition is dominated by quartz and alkali feldspar, with minor biotite and opaque iron oxides.

- **Color and Alteration:** Colors range from greenish-grey to purplish and reddish-brown. This coloration results from post-eruptive hydrothermal alteration and subsequent coastal chemical weathering, which oxidized ferrous iron into reddish ferric oxides.
- **Jointing and Fracture Patterns:** The outcrop is dissected by cooling joints and tectonic fractures, creating angular, blocky rock masses. Sea waves and salt spray continuously attack these fractures, widening cracks through physical and chemical weathering.

Figure 10: Weathered rhyolite at the coast. The outcrop shows angular blocky jointing and oxidized surface alteration.

The rhyolite outcrops provide clear field evidence of past felsic volcanic activity in southern Vietnam, illustrating how high-temperature volcanic rocks are progressively modified by surface weathering and coastal erosion.

2.4 Granitic Intrusion, Diabase Dykes, and Cross-cutting Relationships

Intrusive (plutonic) igneous rocks form when magma cools and crystallizes slowly underground within the Earth's crust. Because cooling takes thousands to millions of years, mineral crystals have sufficient

time to grow large, producing a visible, interlocking coarse-grained (phaneritic) texture.

2.4.1 Granitic Intrusion

At Stop 6 (Mui Ke Ga, Binh Thuan Province), students examined a massive granitic intrusion belonging to the Late Cretaceous Deo Ca Complex:

- **Texture:** Holocrystalline, phaneritic texture with interlocking crystals 3–8 mm across, formed by slow crystallization at depth.
- **Mineral Composition:**
  - **Quartz (**25–35%**):** Glassy, translucent smoky-gray grains.
  - **Potassium Feldspar (**40–50%**):** Subhedral pinkish to milky-white orthoclase crystals, imparting a pinkish-grey tone to the granite.
  - **Plagioclase (**15–20%**):** White tabular crystals.
  - **Biotite Mica (**5–10%**):** Shiny black platy flakes with perfect cleavage.
- **Structural Joints and Marine Abrasion:** The granite body is cut by orthogonal tectonic joint sets and sub-horizontal exfoliation (unloading) fractures. Marine wave action exploits these joints, carving wave-cut platforms, sea cliffs, and rounded boulders known as corestones and tors.

2.4.2 Diabase Dykes

Diabase (also known as dolerite) is a mafic intrusive rock that is the subvolcanic equivalent of basalt. It forms when basaltic magma intrudes into fractures within pre-existing rocks at shallow to intermediate depths:

- **Texture and Color:** Medium- to fine-grained texture (finer than granite due to faster cooling in narrow fractures). It exhibits a dark greenish-grey to dark brownish-red color.
- **Mineral Composition:** Composed mainly of calcic plagioclase laths and augite pyroxene, with minor iron oxides.
- **Occurrence:** Observed cutting through the host rock as tabular, steep-dipping dykes with sharp intrusive contact margins.

Figure 11: Diabase dyke and host rock at Mui Ke Ga. The dark dyke cuts across lighter host rock, showing a distinct geological contact.

2.4.3 Cross-cutting Relationship of Intrusive Igneous Rocks

A fundamental principle of relative dating in geology is the Principle of Cross-Cutting Relationships, established by James Hutton:

> *"Any geological body or fault that cuts across or penetrates another rock unit must be younger than the rock unit it cuts."*

At Mui Ke Ga, dark diabase dykes slice cleanly through the light-colored host granite:

1. The host granite formed first through deep-seated crystallization of silica-rich magma during the Cretaceous.
2. After the granite had solidified and cooled, tectonic stresses generated fracture systems within the crystalline pluton.
3. Later, mafic magma ascended from the mantle and intruded along these fractures, solidifying to form the diabase dykes.
4. Because the diabase cuts through the granite, the diabase is definitively established as the younger magmatic event.

Figure 12: Cross-cutting relationship at Mui Ke Ga. A younger dark dyke cleanly transects the pre-existing jointed granitic host rock.

This field relationship demonstrates how geologists determine the chronological sequence of geological events directly from geometric contact relationships in the field.

2.4.4 Height Estimation of the Red Sand Escarpment Using a Human Scale

In geological fieldwork, establishing a physical scale reference is essential for quantitative morphological analysis and engineering slope evaluation. In Figure 13, a student with my group with a height of 1.78 m stands barefoot at the base of the red sand talus cone, providing an exact human scale reference for the formation:

Figure 13: Red and white sand escarpment at Suoi Tien. Using the student in the foreground (1.78 m tall) as a visual scale reference, the total vertical height of the cliff from the stream base to the vegetated crest is estimated at approximately 7 to 7.5 meters.

By analyzing the relative vertical proportions between the student (1.78 m) and the geological outcrop:

1. **Talus Cone Base (Lower Zone):** The sloping red sand cone from the stream channel up to the student's feet measures approximately 1.5–2.0 m in vertical height.
2. **Middle Vertical White Sand Face:** The vertical white cliff face exposed directly behind and above the student stands approximately 3.0–3.5 m tall (roughly 1.8 to 2.0 times the student's height).
3. **Upper Red Sand Crest (Top Cap):** The uppermost red sand layer extending to the top ridge with natural shrubs adds approximately 1.5–2.0 m of additional vertical relief.
4. **Total Estimated Formation Height:** Summing these geological layers, the total exposed height of the red sand escarpment from the active stream bed to the top crest is estimated at approximately 7 to 7.5 meters (approximately 3.9–4.2 times the height of the 1.78 m student).

2.4.5 Geomorphological and Slope Stability Implications

The ability of this weakly consolidated sand formation to maintain steep vertical walls up to 7–7.5 m in height is attributed to the apparent cohesion provided by capillary soil moisture and the cementing effect of the fine ferruginous clay matrix. However, continuous lateral undercutting by the stream at the base combined with intense monsoonal rainwash continuously undermines slope stability, triggering periodic gravitational slumping and driving rapid micro-canyon evolution.

2.5 Measuring the Orientation of Fractures with a Geological Compass

Besides identifying rock types, structural geology requires recording the orientation of planar features such as joints, faults, and bedding. During the fieldtrip, the group used a geological compass (Brunton-type compass with a clinometer) to measure the orientation of fracture planes in the rock [16]. Measurements were taken at two outcrops: Bai Sau (Vung Tau) and Mui Ke Ga. The orientation of a plane is described by three values:

- **Strike:** the direction of the horizontal line formed where the plane meets an imaginary horizontal surface, measured as an azimuth from North.
- **Dip direction:** the azimuth of the line of steepest descent on the plane, which is perpendicular to the strike (the strike differs from the dip direction by 90°).
- **Dip angle:** the angle between the plane and the horizontal, measured downward from 0° (horizontal) to 90° (vertical).

All directions are measured clockwise from geographic North (0°). To measure strike and dip direction, the compass is held level (using the bull's-eye level) against the fracture surface and the needle is read after it settles. The dip angle is measured by turning the compass on its edge, with its long side placed along the line of steepest descent, and reading the clinometer after leveling it [16].

2.5.1 Measurements at Bai Sau (Vung Tau)

At Bai Sau, a fracture plane in the dark, fractured volcanic rock was measured. The compass readings are shown in Figure 14.

(a) Strike: 035° (b) Dip direction: 310° (c) Dip angle: 90°

Figure 14: Compass measurements of a fracture plane at Bai Sau: (a) strike, (b) dip direction, and (c) dip angle.

The measured values are a strike of 035° (N35°E), a dip direction of 310°, and a dip angle of 90°. The dip direction is almost perpendicular to the strike (the expected value is 305° or 125°), so the small difference of about 5° can be explained by reading error and by the roughness of the rock surface. The dip angle of 90° means that the fracture is vertical. For a vertical plane the dip direction is not unique, because the plane dips equally to both sides, so the dip direction is determined only by the side of the plane on which the compass is placed. The fracture can therefore be described as a vertical joint striking NE–SW.

2.5.2 Measurements at Mui Ke Ga

At Mui Ke Ga, the same three measurements were repeated on a fracture surface in the granite, as shown in Figure 15.

(a) Strike: 225° (b) Dip direction: 315° (c) Dip angle: 85°

Figure 15: Compass measurements of a fracture plane in the granite at Mui Ke Ga: (a) strike, (b) dip direction, and (c) dip angle.

The measured values are a strike of 225° (SW–NE direction), a dip direction of 315°, and a dip angle of 85°. The dip direction is exactly 90° from the strike ($225° + 90° = 315°$), which confirms that the three readings are consistent. The fracture is nearly vertical, dipping steeply toward the northwest.

2.5.3 Comparison and Geological Significance

Table 3: Compass measurements of fracture planes

Location	Strike	Dip direction	Dip angle	Type of plane
Bai Sau (Vung Tau)	035°	310°	90°	Vertical joint
Mui Ke Ga	225°	315°	85°	Near-vertical joint

The two outcrops are in different rock types, but both fractures are vertical or almost vertical, with a strike in the NE–SW direction. A strike line can be read in either of two opposite directions, so 225° is equivalent to 045°, which differs from the Bai Sau strike (035°) by only about 10°. This similarity suggests that comparable stress directions affected both areas, although two measurements are too few to draw a regional conclusion. Steep joints like these are formed when the rock cools and contracts, or when it is later stressed by regional tectonic movements, and they provide pathways for water and plant

roots. This is important at Mui Ke Ga, where fractures in the granite guide weathering and wave erosion, and where the diabase dykes described above were emplaced along similar fracture systems. Orientation measurements such as these are therefore the basis for interpreting rock structure, slope stability, and fluid flow in the field.

2.6 Summary of Geosphere Observations

The geosphere observations during the fieldtrip provided students with direct evidence of the cyclical nature of Earth's internal and surface processes. The main conclusions from the five studied topics are synthesized below:

Table 4: Summary of geosphere field observations. Key processes recorded across the visited stops.

Geological Topic	**Visited Location**	**Observed Rock / Material**	**Primary Geological Process / Principle**
Weathering in Laterite	Stop 1: Ho Da (VNU-HCM Lake)	Andesite bedrock, residual laterite mantle (đá ong)	Tropical chemical leaching of silica and alkaline cations; in-situ concentration of Fe-Al sesquioxides.
Erosion & Coastal Sedimentation	Stop 2: Bai Sau (Vung Tau) Stop 5: Cua Ho Tram – Loc An	Quartz sand, ilmenite heavy minerals, sand dunes, rip-rap seawall	Marine wave swash-backwash sorting, hydraulic placer concentration, coastal erosion, aeolian wind ripples.
Magma Eruption Processes & Rhyolite	Stop 2: Bai Sau (Vung Tau) Stop 4: Gia Quy Quarry	Nha Trang Formation rhyolite, Xuan Loc Formation vesicular basalt	Magma viscosity control by silica (SiO_2) content; rapid surface cooling; volcanic degassing vesicles.
Granitic Intrusion & Cross-cutting	Stop 6: Mui Ke Ga (Ke Ga Cape)	Deo Ca Complex coarse-grained granite, intrusive diabase dykes	Deep plutonic crystallization (phaneritic texture); James Hutton's Principle of Cross-Cutting Relationships.
Fracture Orientation (Compass)	Stop 2: Bai Sau (Vung Tau) Stop 6: Mui Ke Ga	Vertical to near-vertical joints in volcanic rock and granite	Strike, dip direction, and dip angle measured with a geological compass: $035°/310°/90°$ and $225°/315°/85°$.
Deep Petroleum Geosphere	Stop 3: Vietsovpetro (Vung Tau)	Fractured granitic basement, Oligocene syn-rift shales	Cuu Long rift basin evolution; secondary fracture porosity (8–15%) hosting commercial hydrocarbons.
Fluvial-Aeolian Micro-Canyon	Stop 7: Suoi Tien (Lam Dong)	Semi-lithified red sand-clay strata, white quartz sand	Stream basal bank undercutting, mass wasting, rainwash rill fluting, and badland pinnacles (fairy chimneys).

In addition, the observations demonstrate that the geosphere does not operate in isolation, but constantly interacts with the other Earth spheres:

- **Atmosphere $\longleftrightarrow$ Geosphere:** Atmospheric monsoonal precipitation and high temperatures accelerate chemical lateritization, while onshore winds transport loose sand to build coastal dunes and shape asymmetric ripples.

- **Hydrosphere $\longleftrightarrow$ Geosphere:** Ocean wave kinetic energy erodes sea cliffs and wave-cut platforms on rocky granitic coasts, while wave winnowing sorts beach sand and concentrates heavy ilmenite placers on sandy shores.
- **Biosphere $\longleftrightarrow$ Geosphere:** Plant roots penetrate joint fractures in granite and laterite, exerting wedging pressures that accelerate mechanical weathering, while decaying vegetation releases organic acids that enhance mineral leaching.

Overall, these field investigations helped students connect classroom theory with real geological outcrops, illustrating how internal magmatic forces build new rock formations while external weathering and coastal forces continuously reshape the Earth's surface

3 Hydrosphere

3.1 Surface Water and Stream Processes at Suoi Tien (Fairy Stream)

Suoi Tien (Fairy Stream) represents a unique surface water system flowing through a micro-canyon terrain shaped by wind and water erosion. The shallow stream is characterized by a steady, low-velocity laminar-to-turbulent flow that constantly transports fine sand particles down to coastal estuaries.

As surface water flows along the channel, it continuously undercuts the base of the surrounding sand and clay cliffs. This lateral stream erosion, combined with rainfall runoff, causes small-scale slope instability and mass wasting along the stream margins. The interaction between surface water and weathered sand formations demonstrates active fluvial-aeolian erosion and sediment transport within a coastal stream environment.

Figure 16: Stream erosion at Suoi Tien. Surface water flow has undercut the sand cliffs.

3.2 Groundwater Dynamics in Coastal Sand Dune Systems (Ho Tram & Coastal Dunes)

Coastal sand dunes, such as those observed around Ho Tram and adjacent coastal stretches, play a critical role in local hydrogeology. Because loose dune sand possesses high porosity and permeability, rainwater easily infiltrates directly into the subsurface with minimal surface runoff, establishing these dune fields as effective natural groundwater recharge zones.

Rainwater penetrating through the sandy layers forms a shallow freshwater lens above denser saline water. During field observations, local groundwater wells and natural seepages along dune depressions demonstrated the availability of accessible freshwater in sandy environments. However, these shallow aquifers remain highly sensitive to over-extraction and sea-level fluctuations, which can trigger saltwater intrusion into the fresh groundwater zone.

Figure 17: Coastal sand dunes and groundwater recharge. The aeolian dune surface at Ho Tram acts as a groundwater recharge zone.

3.3 Coastal Erosion, Wave Action, and Marine Processes (Back Beach, Mui Ke Ga, & Ho Tram Estuary)

Coastal hydrological processes were investigated across multiple distinct coastal environments, including Back Beach (Vung Tau), Mui Ke Ga, and the Ho Tram river mouth / coastal inlet (Cua Ho Tram).

At Back Beach and Cua Ho Tram, continuous wave action and tidal currents drive intense coastal sediment transport. High-energy waves strike the shoreline, creating steep beach scarps and eroding sandy shorelines, while longshore currents carry loose sand along the coast to form depositional bars near river inlets.

At Mui Ke Ga, where rocky headlands intersect the ocean, wave energy is concentrated directly against bedrock exposures. Mechanical wave battering and hydraulic fracturing constantly weather and erode the granitic headland structures, creating wave-cut platforms and sea cliffs. The contrasting dynamics between the sandy beaches of Ho Tram / Vung Tau and the rocky exposures of Mui Ke Ga highlight how coastlines respond differently to ocean hydrodynamics based on local geology.

3.4 Impact of Mining Pit Water Retention at Ho Da and Gia Quy Quarry

The hydrosphere also interacts closely with human engineering and resource extraction activities, as observed at Ho Da (VNU-HCM Quarry lake) and Gia Quy Quarry.

At Ho Da, past quarrying deepened the basin below the regional water table, allowing groundwater seepage and surface runoff to accumulate over time. This created a permanent artificial quarry lake. Similarly, open-pit operations at Gia Quy Quarry alter local surface runoff paths and groundwater drainage. These sites demonstrate how anthropogenic excavations interrupt natural hydrological flow paths and create standing water bodies within former mining pits.

Figure 18: Coastal erosion and wave action. An eroded sandy beach scarp shows the effect of waves along the coast.

Figure 19: Waves acting on granite at Mui Ke Ga. Wave action is visible against the rocky headland.

Figure 20: Water accumulation at Ho Da Quarry. Surface water and groundwater fill the former quarry pit.

4 Biosphere

The biosphere includes all living organisms and the environments in which they live. During the fieldtrip, plants were the most visible part of the biosphere. At every stop, vegetation was found growing on rock, sand, or sediment, and in each case it was closely tied to the local geology and water conditions. This section describes these observations by location and shows how the biosphere interacts with the geosphere and hydrosphere.

4.1 Pioneer Vegetation and Biological Weathering at Ho Da (VNU-HCM Lake)

Biological weathering is the breakdown of rocks caused by living organisms such as plants, roots, microorganisms, and animals. Among these agents, plant roots are one of the most common causes of biological weathering in exposed rock areas. When seeds fall into small cracks or fractures in rocks, they can germinate if there is enough moisture and soil material. As the plant grows, its roots gradually extend deeper into the fractures and exert pressure on the surrounding rock. This pressure slowly widens the cracks and weakens the rock structure, so the rock may split into smaller blocks [15].

At Ho Da, the old quarry pit is now filled with water, and its rim is covered by tall grass, shrubs, and small trees growing on the fractured andesite and weathered laterite soil. Because quarrying removed the original soil cover, these plants are pioneer species: they are the first organisms to colonize the bare rock surface. Their presence shows that weathered material and moisture have accumulated in the fractures and on the rock surface, which supports plant growth.

This example shows the connection between the biosphere and geosphere. Roots and organic acids help weaken the rock, and the new cracks allow more water and air to enter, which increases chemical weathering and physical breakdown. Over time, the plant cover also stabilizes the loose soil on the pit slopes and reduces surface erosion, which supports the natural restoration of the post-mining landscape.

4.2 Plant Colonization of Basalt and Pozzolan Surfaces at Gia Quy Quarry

Gia Quy Quarry is an active open-pit site where fresh basalt and pozzolan are exposed. The rock surface here is disturbed more recently than at Ho Da, yet plants were still found in many places. Small herba-

Figure 21: Vegetation colonising the Ho Da quarry rim. Grass and shrubs grow around the former quarry pit, with water in the background.

ceous plants grew between broken basalt blocks and in the gravel on the quarry floor, and grass covered parts of the weathered slopes and benches of the pit wall.

Basalt weathers into clay-rich, nutrient-bearing soil that holds more water than fresh rock. Therefore, the plants appear first where weathered material, fine soil, or pozzolan fragments have gathered between blocks. The roots of these plants enter the spaces between fragments, hold the fine particles in place, and add organic matter as they die and decay. In this way vegetation slowly changes loose quarry waste into a thin soil layer. The same plants also open the rock fractures further, so the biosphere both builds soil and breaks down rock at this site.

4.3 Coastal Vegetation at Bai Sau and Cua Ho Tram – Loc An

Coastal vegetation was observed at both Bai Sau (Vung Tau) and the Cua Ho Tram – Loc An coast. At the back of the beach, sandy ground was covered by casuarina trees, low shrubs, and creeping plants. Casuarina has a deep, spreading root system and tolerates strong wind, salt spray, and dry sandy soil, and it is widely used for dune stabilization and windbreaks on sandy coasts [17].

At Cua Ho Tram – Loc An, creeping plants with round leaves and purple flowers (probably beach morning glory) covered the sand behind the shoreline, and green shrubs grew next to the concrete tetrapods and the rip-rap seawall. Creeping plants spread along the ground and bind loose sand with their stems and roots. Together with the shrubs, they reduce the movement of sand by wind and by waves on the upper beach. This is the same sand that forms the aeolian dunes and ripples described in the geosphere section.

These observations show how vegetation and engineering structures work together for coastal protection. The seawall and tetrapods reduce wave energy on the shoreline, while the plants hold the sand behind them and limit wind erosion. At river mouths and estuaries such as Cua Ho Tram, mangrove forests do a similar job in muddy, brackish environments. Their dense roots trap fine sediment and slow down waves and currents, although the protection depends on the local conditions [18]. Mangroves also provide habitats for fish, crabs, birds, and microorganisms. Mangroves were not the main focus of our observations, but they show the same principle that we saw on the sandy coast: living plants stabilize sediments and protect the shoreline.

Figure 22: Pioneer plants in Gia Quy Quarry. Small plants grow between broken basalt fragments.

Figure 23: Grass growing on Gia Quy Quarry slopes. The weathered slope shows early soil stabilization.

Figure 24: Coastal vegetation at Bai Sau. Casuarina trees and shrubs grow behind the beach in Vung Tau.

Figure 25: Creeping vegetation and shrubs at Loc An. Left: creeping plants with purple flowers on sand at Cua Ho Tram – Loc An. Right: shrubs growing beside the concrete tetrapods of the coastal protection structure.

4.4 Vegetation on the Rocky Headland at Mui Ke Ga

Mui Ke Ga is a granite headland where waves strike directly against bedrock. Even in this exposed place, shrubs and low vegetation cover the top of the rocky islets and the soil pockets between the granite boulders, while the rock near the waterline remains bare. Plants cannot survive where waves and salt spray act continuously, so the vegetation grows only above the strongest wave zone. This gives a clear boundary between the marine environment of the rock platform and the land ecosystem on the higher ground.

Figure 26: Vegetation on Mui Ke Ga headland. Plants grow on the granite above the zone of wave action.

The plants on the headland grow in thin soil formed by the weathering of granite and depend on rainfall because the rock holds very little water. Their roots enter joints and fractures in the granite, so they also contribute to the breakdown of the rock, in addition to the wave action and chemical weathering described earlier.

4.5 Vegetation and Slope Stability at Suoi Tien (Fairy Stream)

At Suoi Tien, the stream has cut into red and white sand layers and formed steep cliffs. Trees and shrubs grow along the top of these cliffs and along the stream banks. The sand is loose and easily eroded, so vegetation plays an important role here. Roots hold the sand at the top of the slope and reduce erosion by rainfall runoff.

Figure 27: Vegetation on sand cliffs at Suoi Tien. Trees and shrubs grow on top of the red sand cliff.

However, vegetation cannot fully prevent erosion on these steep slopes. In one location, a tree had fallen on the white sand cliff with its roots exposed, probably after the stream undercut the base of the slope and the ground collapsed. This shows that the biosphere is also affected by geological processes: erosion and slope failure can remove plants, and the fallen wood and sediment then enter the stream channel.

4.6 Summary of Biosphere Observations

Table 5 summarizes the biosphere observations made at each location and their relationship with the other Earth spheres.

Overall, the fieldtrip showed that the biosphere cannot be separated from the other Earth systems. Plants break down rock and produce soil, stabilize slopes and coastal sand, and respond directly to water availability and to the type of rock or sediment on which they grow. At the same time, geological processes such as erosion, wave action, and quarrying control where plants can grow and which of them survive.

5 Atmosphere

The coastal route from Vung Tau to Phan Thiet provides an opportunity to relate regional climate to the landscapes studied during the fieldtrip. Rainfall seasonality, atmospheric moisture, and wind help explain the conditions under which exposed rocks weather and loose coastal sediments are reworked.

Figure 28: Fallen tree and slope erosion at Suoi Tien. The fallen tree on the eroded sand cliff shows that slope failure can remove vegetation.

Table 5: Summary of biosphere field observations during the fieldtrip.

Location	**Vegetation observed**	**Role and interaction with other spheres**
Ho Da (VNU-HCM Lake)	Grass, shrubs, and small trees on the quarry rim	Pioneer colonization; root weathering of fractured rock; slope stabilization (biosphere–geosphere)
Bai Sau	Casuarina trees and coastal shrubs	Protection of the back-beach sand from wind and salt spray (biosphere–atmosphere–geosphere)
Gia Quy Quarry	Herbs and grass on basalt and pozzolan	Soil formation from weathered basalt; stabilization of quarry waste (biosphere–geosphere)
Cua Ho Tram – Loc An	Creeping plants and shrubs near the seawall	Binding of sand; support for coastal protection structures (biosphere–hydrosphere–geosphere)
Mui Ke Ga	Shrubs on the granite headland	Colonization above the wave zone; root growth in joints (biosphere–geosphere–hydrosphere)
Suoi Tien	Trees on the sand cliff and stream banks	Slope stabilization by roots; removal of vegetation by bank collapse (biosphere–geosphere–hydrosphere)

5.1 Climatic Characteristics of Vung Tau

Vung Tau has a warm climate throughout the year, with a clear seasonal distribution of rainfall. The climatic data published in QCVN 02:2022/BXD give an annual mean air temperature of 27.1 °C and annual rainfall of approximately 1,419 mm at Vung Tau station. Most rainfall occurs from May to October, while the first months of the year are much drier. Annual mean relative humidity is approximately 80% [1].

This seasonal pattern provides context for the coastal environments visited at Bai Sau and the Loc An coastal site (Cua Ho Tram). Rain supplies moisture to exposed soils and fractured rock, while subsequent drying changes the condition of surface materials. Warm, moist conditions support chemical weathering where water can react with minerals. At sandy sites, alternating wet and dry conditions also influence the availability of loose grains for wind transport. The atmosphere therefore contributes to landscape development through both the supply of water and the movement of air.

Figure 29: Atmospheric conditions at Bai Sau. The coastal setting shows the sky, sandy beach, exposed rocks, and shoreline vegetation.

5.2 Climatic Characteristics of Phan Thiet and Their Landscape Significance

Phan Thiet is also warm throughout the year, but receives less annual rainfall than Vung Tau. The same national dataset reports an annual mean temperature of 26.9 °C and annual rainfall of approximately 1,089 mm at Phan Thiet station [1]. The city's 2023 land-use planning report describes a rainy season from approximately mid-May to late October and a dry season from November to April, together with around 2,600–2,700 hours of sunshine annually [2]. These figures indicate that the regional contrast is particularly evident in rainfall availability, while annual mean temperatures remain similar.

For the sand-rich landscapes around Mui Ne, including the surroundings of Suoi Tien, this climatic setting helps explain the importance of surface drying and wind action. When sand is exposed, sufficiently dry, and not protected by vegetation, wind can move grains by short hops and rolling, producing erosion and deposition. The amount of sand moved depends on wind strength, sediment supply, surface moisture, and vegetation cover [3]. The balance between rainfall, surface drying, and wind exposure therefore helps determine when loose sand is available for transport.

Figure 30: Atmospheric conditions at Mui Ke Ga. A cloudy sky hangs above the rocky shoreline and lighthouse.

6 Natural Resources

The resource-related stops helped us examine how geological materials become useful products and how extraction changes the surrounding environment. Ho Da and Gia Quy provided examples of construction-stone resources at different stages of land use, while Vietsovpetro introduced the engineering systems required to develop offshore petroleum resources. Together, these sites connected resource availability with material quality, technical choices, economic efficiency, and responsibility for the landscape left after extraction. The rock examples discussed below are selected illustrations from the wider survey, including andesite at Ho Da, basalt at Gia Quy, and granite at Mui Ke Ga.

6.1 Construction Stone and the Post-mining Landscape at Ho Da

Ho Da, within the VNU-HCM area, is a former quarry where the remaining excavation has become a lake. The exposed rock, water surface, and vegetation visible around the site show how a landscape continues to change after the principal extraction activity has ended. From a resource perspective, the most important feature is the excavation itself: construction stone has been removed, leaving a landform whose future use must account for the geometry of the remaining rock faces and the presence of water.

At Ho Da, our field notes identify andesite as one of the rocks examined. This provides a specific example of the geological material exposed at the former quarry [4]. Removing stone changes the ground surface and the setting for water and vegetation, so the resulting landscape forms part of the assessment of resource development.

The usefulness of construction stone depends on material quality as well as the quantity available. Competent igneous rock can supply aggregates, but weathered, fractured, porous, or altered portions of a deposit may behave differently from fresh rock. Geological assessment consequently examines variations across an exposure and relates them to the intended use. Properties such as particle size, strength, water absorption, and durability require appropriate evaluation before a material is selected for a particular construction application [5]. The exposures at Ho Da provide a field setting in which to discuss this connection between geological condition and engineering performance.

Figure 31: Post-mining landscape at Ho Da. The site has a water-filled excavation, exposed rock faces, and vegetation around the lake.

Vegetation around the lake shows that plant growth can return to disturbed ground. However, assessing a former quarry also requires examining the remaining highwalls and changes in drainage and groundwater conditions. Research on abandoned quarry highwalls emphasizes that slope assessment must continue to consider conditions after extraction stops [13]. The presence of plants and water is therefore only part of the information needed to evaluate a post-mining landscape.

A rehabilitation plan should link the physical condition of the site to a realistic future use. Possible measures include stabilizing slopes, reshaping disturbed ground where appropriate, managing drainage, and establishing vegetation [7]. For Ho Da, the educational value of exposed rock and the landscape can be considered alongside the management of access and steep edges. The lesson from this stop is that responsible resource use includes both obtaining a useful material and planning for the land after that material has been removed.

6.2 Construction Stone Resources and Quarry Engineering at Gia Quy

The group examined a range of rocks and minerals at Gia Quy; the discussion and photographs below present a few illustrative examples. Our field notes describe basalt, including compact and vesicular varieties, and weathered rock with brown surfaces and fractures [4]. The field photographs show exposed quarry faces, accumulations of broken rock, and a material-handling area. Seeing these features together helped connect the rock mass in an exposure with material that can enter the construction supply chain. The quarry face represents the source, while the detached fragments and handling facilities illustrate subsequent stages in using that source.

Basalt and other suitable igneous rocks can be processed into crushed stone [5]. In the aggregate industry, extracted material is crushed and sorted into sizes appropriate for its application; it may then be used directly or combined with cement or asphalt to produce construction materials [6]. The Gia Quy visit provided a practical setting for understanding why a natural rock resource needs selection and processing before it becomes a usable product.

Figure 32: Exposed quarry face at Gia Quy. Fractured rock and loose fragments are visible at the foot of the exposure.

The fractures visible in the quarry face are relevant to both extraction and the condition of the remaining rock mass. Their influence must be considered together with the shape of the excavation and changes in groundwater and surface drainage. Quarry highwall research connects extraction methods and site conditions with both short-term and long-term slope behaviour [13]. Thus, the economic value of a rock deposit cannot be separated from the engineering work needed to access it.

In a quarry layout, benches, drainage, and routes for equipment help organize extraction and movement of material [7]. These are applications of geological and geotechnical knowledge to an excavation. The exposed face and broken blocks made this connection between resource extraction and excavation design visible at the site.

Two selected specimens illustrate some of the mineralogical features discussed during the visit: an olivine-bearing rock and a white calcite specimen. The first contained green olivine in a dark host rock. Olivine is an iron–magnesium silicate, commonly represented by $(Mg, Fe)_2SiO_4$, and occurs in basalt and other igneous rocks. A separate white specimen was identified as calcite, a calcium-carbonate mineral with the formula $CaCO_3$ [12]. These examples showed a contrast between a silicate mineral within a rock and a carbonate mineral specimen.

The samples helped us distinguish the mineralogical interest of an occurrence from the economic assessment of a deposit. Recognizing an individual mineral adds information about a site, while evaluating a construction-stone resource also requires considering the amount, continuity, and engineering quality of the usable material.

Figure 33: Geological samples and facilities at Gia Quy. The images show an olivine-bearing rock specimen (left), a white calcite specimen (centre), and the material-handling area (right).

The use of construction stone also has environmental costs. Quarrying and material handling can generate dust, noise, sediment-bearing runoff, and changes to land cover. Measures such as controlling dust at transfer points, maintaining haul roads, managing drainage, and progressively rehabilitating disturbed areas address these issues [7]. These measures provide a framework for discussing environmental management at the site.

Gia Quy and Ho Da together demonstrate why resource assessment should include both production and the later condition of the extraction site. Rock quality determines possible uses, excavation design affects access and stability, and environmental management influences the effects on surrounding land and water.

Granite at Mui Ke Ga provides another example from the coastal stops. The rocky shore allowed us to relate this geological material to weathering and marine erosion, processes described in our field notes [4]. Together with andesite at Ho Da and basalt at Gia Quy, it illustrates the variety of rocks encountered during the survey. These examples are a selection from the materials studied, and their potential engineering uses still depend on material quality and appropriate evaluation.

6.3 Petroleum Resources and Offshore Technology at Vietsovpetro

Vietsovpetro was established on 19 June 1981 under an intergovernmental agreement between Vietnam and the Soviet Union. Its first crude oil was produced from Bach Ho field on 26 June 1986 [8]. These milestones provided historical context for the organization introduced during the visit. The development of petroleum resources involves establishing the technical capability to investigate the subsurface, construct wells, and support operations offshore; the exhibitions helped connect that wider activity with the models we could examine directly.

During our visit, the group viewed exhibition models and listened to explanations of the organization's activities. The offshore-field model showed multiple platforms, vessels, and connecting routes. Its main educational value was the arrangement of these components: a petroleum field depends on a coordinated system for drilling, production, material movement, and support.

The model helped us distinguish the resource beneath the seabed from the infrastructure needed to develop it. A platform is one part of that system, while other facilities and vessels contribute different functions. Following these relationships on the model helped us study the connection between Earth Science and industrial activity during the onshore exhibition visit.

Figure 34: Offshore-field model at Vietsovpetro. The group listens to an explanation beside the model.

6.3.1 Offshore Structures and Their Engineering Role

The exhibits included a model labelled TAM DAO 02 and a platform model with a steel jacket supporting deck facilities. Vietsovpetro identifies Tam Dao 02 as a jack-up drilling rig [9]. In the model, the drilling tower, working areas, and long supporting legs could be considered together. The adjacent platform model showed a different structural arrangement, with a braced steel framework beneath the deck. Comparing the exhibits helped us recognize that offshore development involves several types of structure with different roles.

The supporting legs also provide a connection to geotechnical engineering. A jack-up foundation must be assessed against the seabed geometry and the soil or rock conditions beneath the footing locations. Differences between those locations can affect the foundation response, so site surveys and subsurface investigations form part of the engineering assessment [14]. This relationship explains why understanding the ground remains important even when the main activity is taking place at sea.

The exhibition therefore illustrated two applications of Earth Science: knowledge used to develop a subsurface resource, and knowledge of seabed conditions used to support the structures that make development possible.

Figure 35: TAM DAO 02 and offshore platform models. The exhibition shows TAM DAO 02 (left) and an offshore platform with a steel jacket and deck facilities (right).

6.3.2 Drilling Technology and Economic Efficiency

One point introduced during the visit concerned the economic benefit of wear-resistant diamond cutting elements. This gave us an example of how a change in material technology can influence the cost of accessing a geological resource. The relevant tool is the drill bit, located at the bottom of the drillstring, where it cuts or breaks the formation.

Polycrystalline diamond compact (PDC) bits use synthetic diamond cutting elements mounted on a supporting body. That body may itself be steel or a tungsten-carbide matrix. The diamond is used at the cutting surface, where resistance to wear is important. PDC cutters remove rock by a shearing action, while roller-cone bits use rolling cutting structures to break the formation [10]. The tools illustrate how cutting materials and bit design affect rock removal.

A study by Vietsovpetro researchers reports that PDC bits were first used at the company's wells in 1994. It describes continued improvements to cutter materials, bit geometry, and compatibility with drilling systems for the Tertiary sedimentary formations of Bach Ho and Rong fields [11]. This provides technical support for the improvement in drilling efficiency discussed during the visit.

The economic reasoning is straightforward. When a bit maintains useful cutting performance for a

longer interval, more metres can be drilled before replacement becomes necessary. Changing a worn bit involves withdrawing the drillstring and returning the replacement assembly to the bottom of the well. Reducing such interruptions can save operating time. Improved drilling speed and longer useful intervals can therefore contribute to a lower overall well-construction cost [10, 11].

The economic comparison must include the complete drilling operation: the bit, equipment time, drilling progress, and replacement work. A wear-resistant cutter is useful when its performance suits the rock and operating conditions. The Vietsovpetro study similarly emphasizes selecting and improving bits for particular well structures and geological conditions [11]. The explanation we received highlights potential savings in total drilling cost through improved tool performance.

Across the three resource-related stops, the main connection was between geology and decisions about use. At a quarry, those decisions concern material selection, excavation, and the future landscape. At an offshore petroleum operation, they also concern the design of structures and the performance of drilling tools. The fieldtrip helped us see how technical knowledge affects both the economic value obtained from a resource and the responsibilities associated with its development.

7 Conclusion

This field trip was a truly unforgettable experience for all members of our group. It was not only a journey to observe rocks, landforms, coastal environments, and natural resources, but also a meaningful opportunity for us to connect classroom knowledge with real field observations. Through each stop of the trip, we learned how Earth Science concepts appear in the real world, from weathering, erosion, and igneous rocks to groundwater, coastal processes, mangrove forests, atmospheric hazards, and natural resource exploitation.

During the fieldtrip, our group had the opportunity to learn about different types of rocks, including intrusive igneous rocks and extrusive volcanic rocks. We observed how magma-related processes can form rocks such as granite, diabase, basalt, and rhyolite, and how these rocks are later modified by weathering, erosion, and human activities. These observations helped us understand geological processes more clearly than learning from textbooks alone.

The fieldtrip also allowed us to apply practical field skills in real situations. We used knowledge of field observation, sample collection, note-taking, photography, and geological description directly at outcrops. In particular, we had the chance to apply compass-related knowledge in the field, which helped us understand how basic field equipment can support geological observation and measurement. These experiences made Earth Science more realistic, practical, and memorable for us.

Every group member had the chance to participate in field activities such as observing outcrops, collecting samples, taking photographs, recording notes, and discussing geological features. These activities helped us realize that Earth Science is not only about theories in lectures, but also about careful observation, teamwork, patience, and curiosity in the field. The trip also allowed us to become closer as a group, as we worked together, supported one another, and shared many memorable moments throughout the journey.

We would like to express our deepest gratitude to Dr. Nguyen Thu Thuy for her dedication, enthusiasm, and tireless teaching throughout the fieldtrip. Despite the long travel schedule and challenging outdoor conditions, she continuously explained the geological features at each site with great energy and passion. Her guidance helped us better understand the meaning of each observation and inspired us to appreciate Earth Science more deeply.

We would also like to sincerely thank MSc. Le Thanh Phong and all the lecturers who organized this fieldtrip. Their careful preparation, guidance, and support created a valuable learning environment for students. Thanks to their efforts, we were able to visit important geological, coastal, and industrial sites and gain practical knowledge that would be difficult to obtain in a classroom setting alone.

In addition, our group would like to thank the supporting staff members and especially Mr. Ngo Tan Tien for assisting our group and the whole fieldtrip team during the journey. Their support helped the trip run smoothly and safely, allowing students to focus on learning, observing, and completing field tasks.

Overall, this fieldtrip will remain a precious and unforgettable memory for all members of our group. It gave us not only academic knowledge, but also practical field experience, friendship, teamwork, gratitude, and respect for nature. We believe that this fieldtrip was more than an academic activity; it was a journey of learning, appreciation, and memories that will stay with us for a long time.

References

[1] Ministry of Construction. *QCVN 02:2022/BXD: National Technical Regulation on Natural Physical and Climatic Data for Construction*. Tables A.2, A.10, and A.25, 2022. https://datafiles.chinhphu.vn/cpp/files/vbpq/2022/10/02-bxd.pdf.

[2] People's Committee of Phan Thiet City. *Explanatory Report on the 2023 Land-Use Plan of Phan Thiet City*. Section 1.1.3: Climate, p. 8, 2023. https://storage-vnportal.vnpt.vn/btn-ubnd/4776/CongkhaiKHSDD2023/BAO%20CAO%20KHSDD%202023%20PHAN%20THIET%20NGAY%2012062023.pdf.

[3] National Park Service. *Monitoring Aeolian Features and Processes*. https://www.nps.gov/articles/aeolian.htm.

[4] Fieldtrip group. *Field notebook (Nhat ky)*. Entries for Ho Da, Gia Quy, and Mui Ke Ga, 30–31 October 2026. Unpublished field notes supplied with this report.

[5] W. H. Langer and D. H. Knepper, Jr. *Geologic Characterization of Natural Aggregate*. U.S. Geological Survey Open-File Report 95-582, 1995. https://pubs.usgs.gov/of/1995/ofr-95-0582/ofr-95-0582.pdf.

[6] T. I. Nelson and W. P. Bolen. *Construction Aggregates*. U.S. Geological Survey, 2007. https://www.usgs.gov/publications/construction-aggregates.

[7] International Finance Corporation. *Environmental, Health, and Safety Guidelines for Construction Materials Extraction*. 2007. https://www.ifc.org/content/dam/ifc/doc/2000/2007-construction-materials-extraction-ehs-guidelines-en.pdf.

[8] Vietsovpetro. *History of Formation*. https://www.vietsov.com.vn/_catalogs/masterpage/lichsu.aspx.

[9] Vietsovpetro. *Facilities and Equipment*. https://www.vietsov.com.vn/_catalogs/masterpage/cosovatchat.aspx.

[10] SLB. *The Defining Series: Bits*. 2016. https://www.slb.com/resource-library/oilfield-review/defining-series/defining-bits.

[11] Đặng Của, Bùi Lê Trọng Hóa, Tạ Ngọc Ánh, Tạ Văn Thịnh, and Nguyễn Văn Cường. *Improving the Structure of Polycrystalline Diamond Compact Drill Bit to Enhance Efficiency When Drilling Through Tertiary Sediments of Bach Ho and Rong Fields, Cuu Long Basin*. Petrovietnam Journal, no. 3, pp. 57–65, 2024. https://doi.org/10.47800/PVSI.2024.03-06.

[12] National Park Service. *Minerals*. Sections on olivine and calcite. https://www.nps.gov/subjects/geology/minerals.htm.

[13] C. Haycocks, G. Haycocks, and Y. Zhou. *Long-term Stability of Abandoned Quarry Highwalls*. 1991. NIOSH Mining Publications. https://stacks.cdc.gov/view/cdc/245526.

[14] L. Kellezi, G. Kudsk, and H. Hofstede. *Jack-up Rig Foundation Design Applying 3D FE Structure–Soil–Interaction Modelling*. Proceedings of the BGA International Conference on Foundations, Dundee, 2008. https://www.geo.dk/media/1184/538_1275_kell_dk_kuds_dk_hofs_nl_070208_rmm.pdf.

[15] S. Earle. *Physical Geology*, 2nd ed., Chapter 5: Weathering and Soil. BCcampus OpenEd. https://opentextbc.ca/physicalgeology2ed/.

[16] S. Earle. *Physical Geology*, section "Measuring Geological Structures". BCcampus OpenEd. https://opentextbc.ca/physicalgeologyh5p/chapter/measuring-geological-structures/.

[17] World Agroforestry (ICRAF). *Casuarina equisetifolia*, Agroforestree Database species profile. https://apps.worldagroforestry.org/treedb2/AFTPDFS/Casuarina_equisetifolia.PDF.

[18] K. B. Gedan, M. L. Kirwan, E. Wolanski, E. B. Barbier, and B. R. Silliman. *The present and future role of coastal wetland vegetation in protecting shorelines: Answering recent challenges to the paradigm*. Climatic Change, vol. 106, no. 1, pp. 7–29, 2011. https://doi.org/10.1007/s10584-010-0003-7.